

ECE2004: Fundamentals of Circuits

Slide Set #3: Ohm's Law, KVL, KCL

Michael Knox



NYU

TANDON SCHOOL
OF ENGINEERING

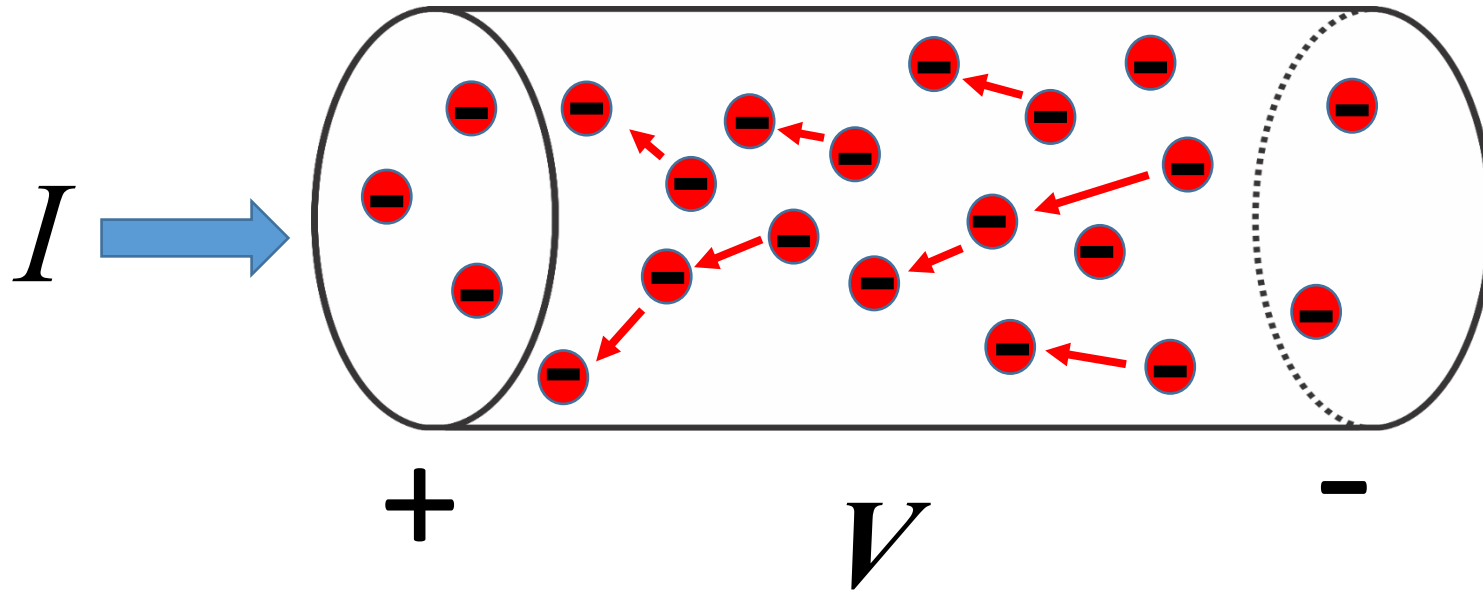
Outline

- Ohm's Law
- Parallel and Series Components
- Kirchhoff's Law
- Ground
- Terminology of Short and Open Circuit



Property of Resistance

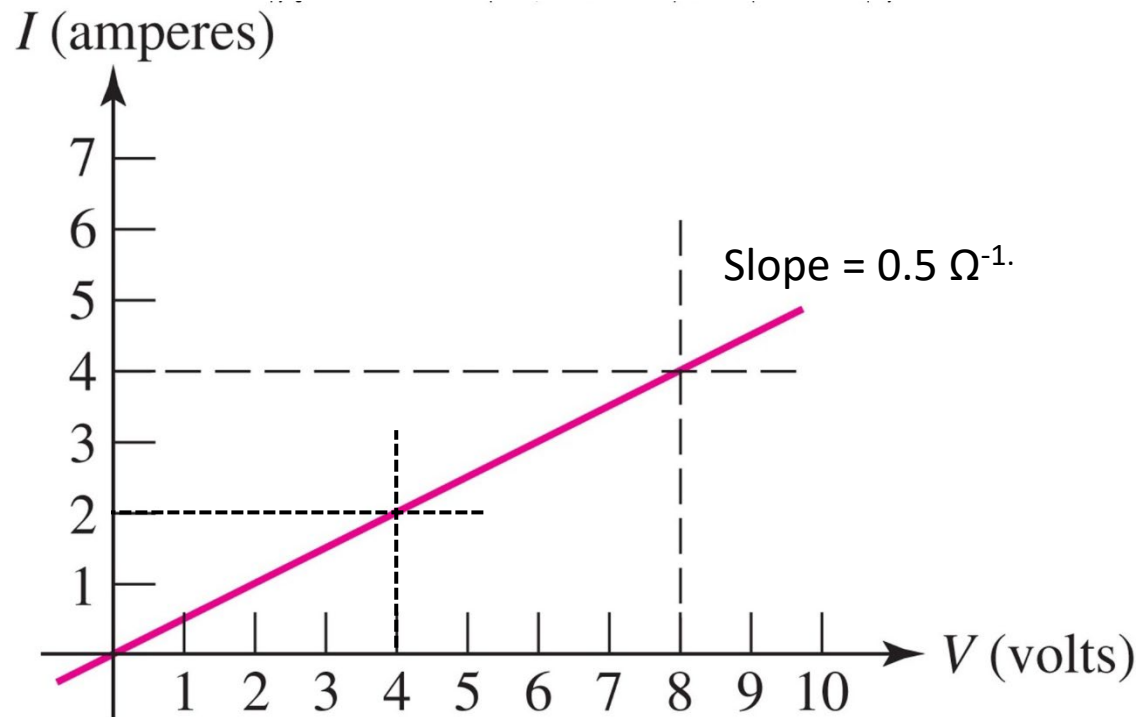
- Certain materials impede the flow of current
- Electrons collide with other electrons, ions, impurities



A voltage difference V across the material drives a current I through it



Ohm's Law



$$R = \frac{8 V}{4 A} = 2 \text{ ohm} = 2 \Omega$$

$$R = \frac{4 V}{2 A} = 2 \text{ ohm} = 2 \Omega$$

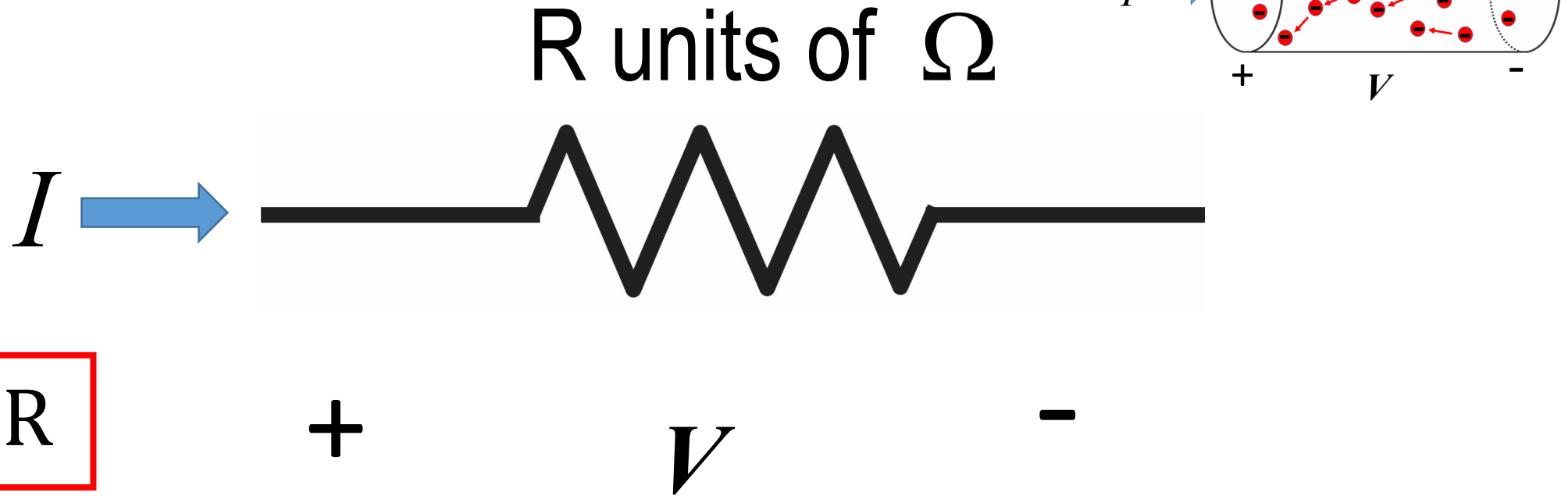
$$V = IR$$

$$R = \frac{V}{I}$$

$$I = \frac{V}{R}$$



Ohm's Law

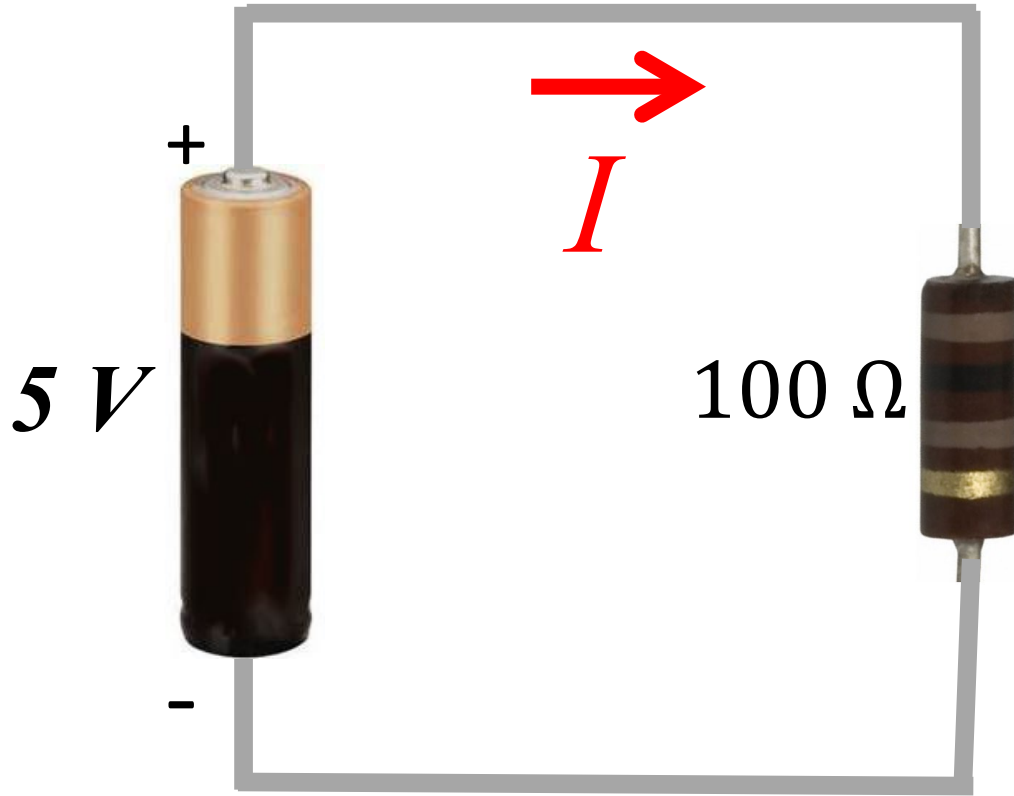


- Current flows in direction of voltage drop
- Resistors only dissipate power

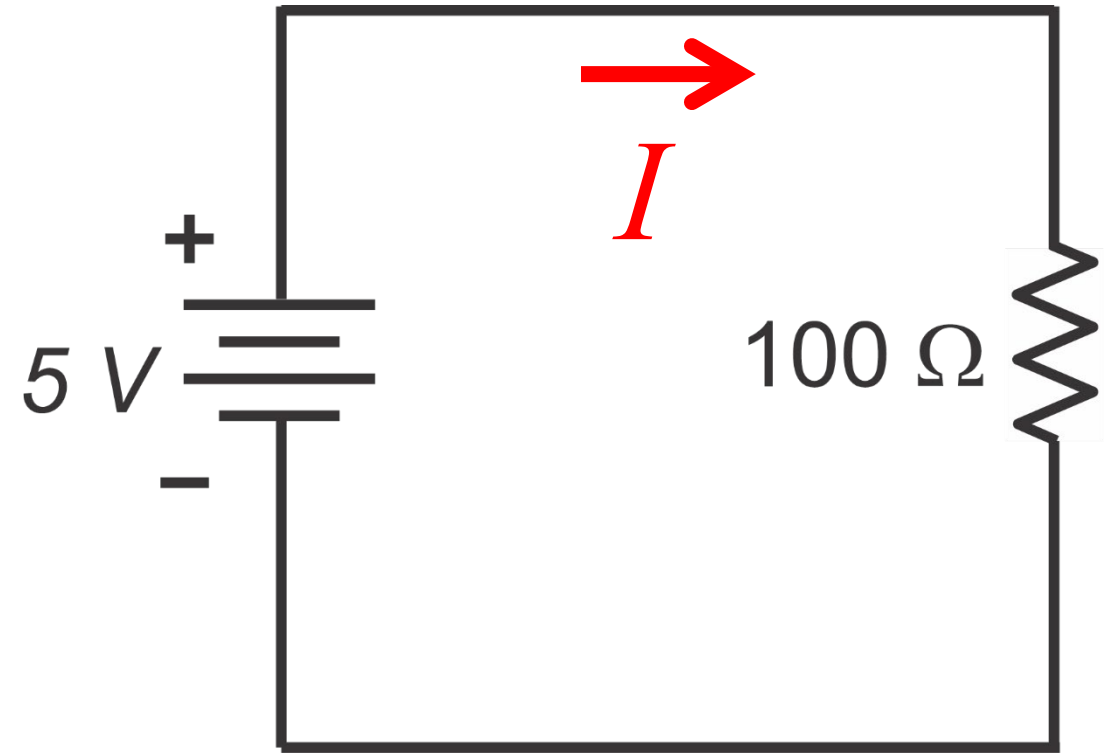


Resistor Connected to Battery

Illustration



Schematic



$$I = \frac{V}{R} = \frac{5V}{100\Omega} = 50 \text{ mA}$$

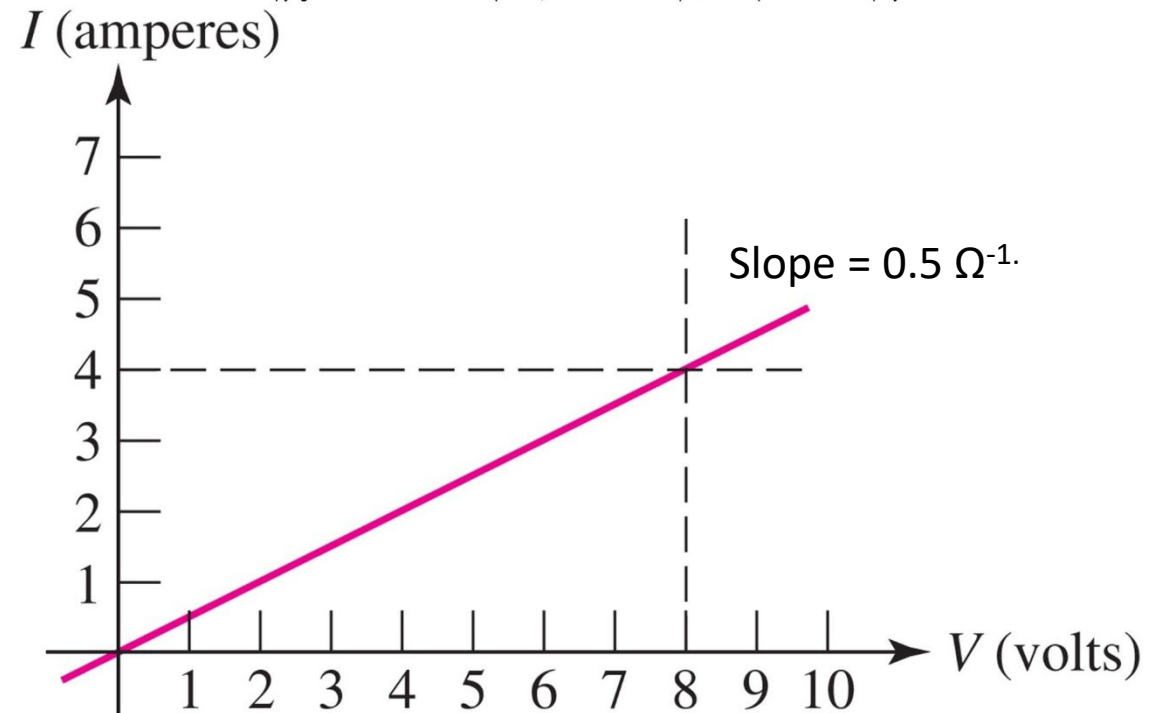


Conductance (G)

$$G = \frac{1}{R} = \frac{I}{V}$$

Units:

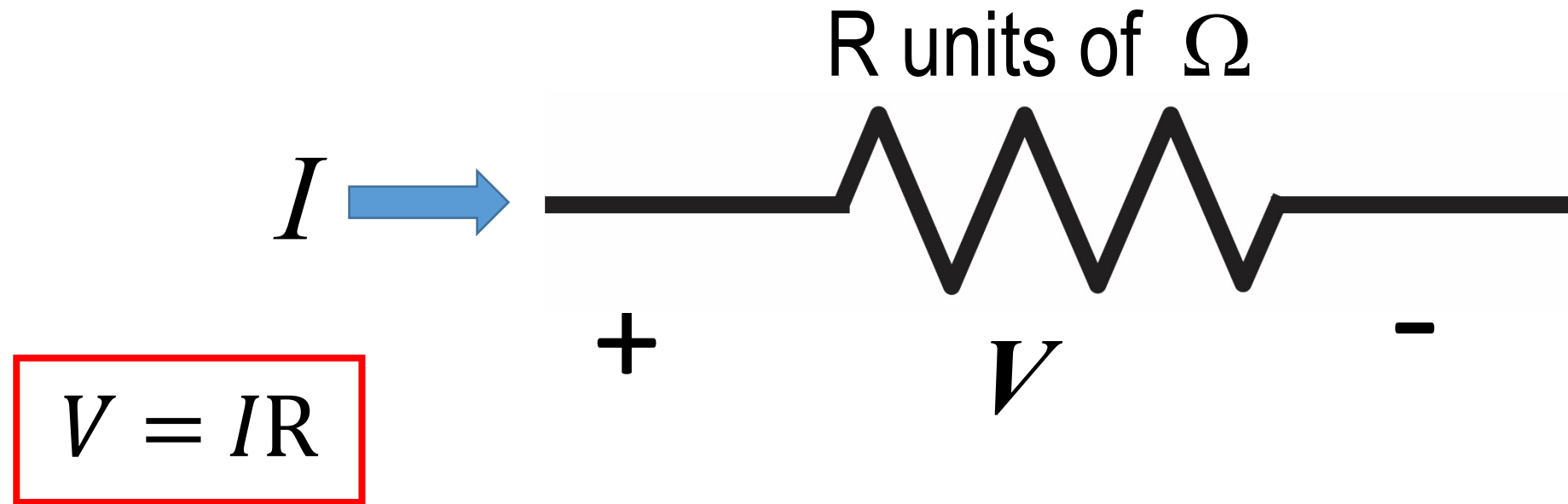
$$\text{Siemens} = \text{mho} = \Omega^{-1}$$



$$R = \frac{8 V}{4 A} = 2 \Omega$$

$$G = \frac{4 A}{8 V} = 0.5 \Omega^{-1} = \text{slope}$$

Resistors Absorb and Dissipate Power

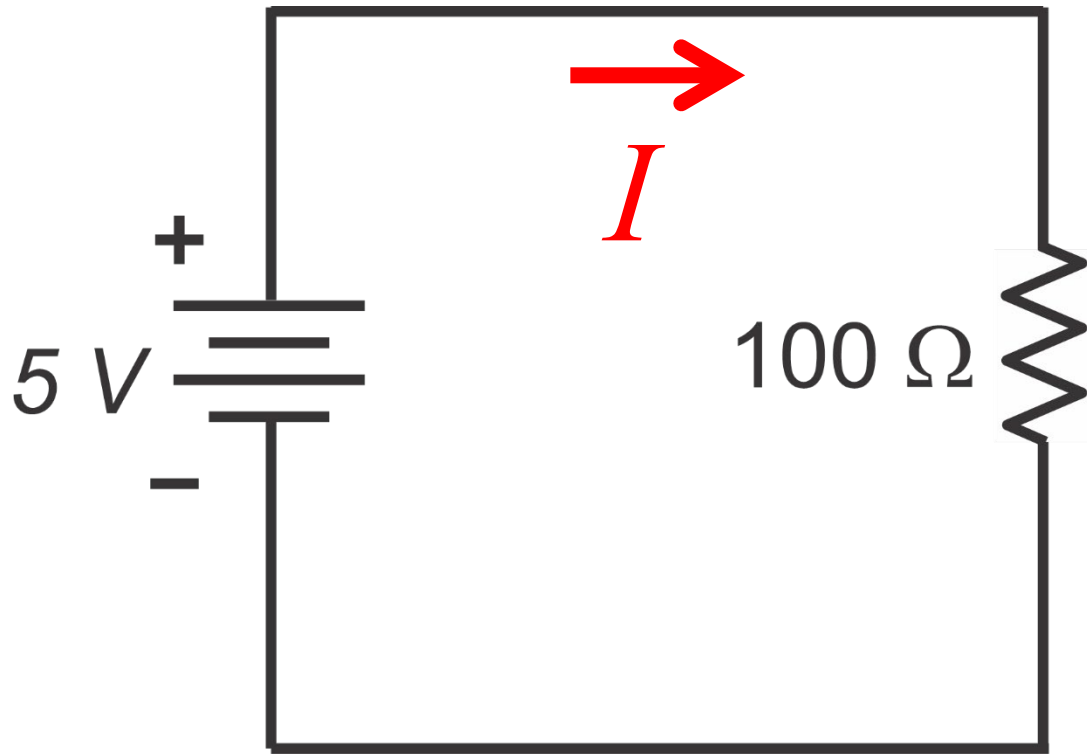


$$P(\text{watts}) = VI = (IR)I = I^2R = \frac{V^2}{R}$$

- Positive power means the resistor is *absorbing* energy
- Power is always positive for a resistor (current enters + terminal) 

Resistors Absorb Power

Schematic



$$I = \frac{V}{R} = \frac{5V}{100\Omega} = 50 \text{ mA}$$

$$P = VI = (5V)(50 \text{ mA})$$

$$P = 0.25 \text{ W} = 250 \text{ mW}$$

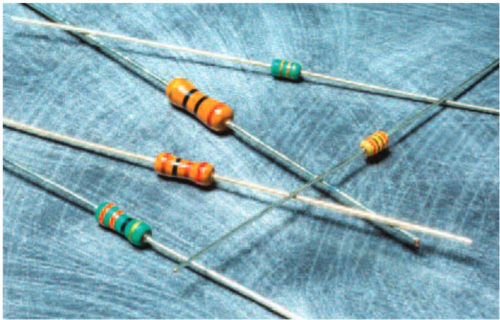
Purchase resistors with

- Ohm value
- % Tolerance
- Power rating (maximum) 🔊

Resistor Datasheet



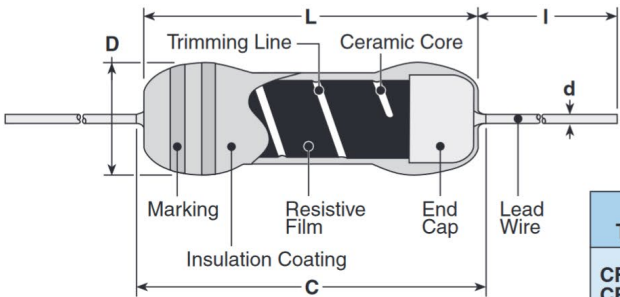
CF
carbon film leaded resistors



features

- Flameproof coating is available (specify “CFP”)
- Reduced body size (specify “CFS/CFPS”)
- Suitable for automatic machine insertion
- Stronger in pulse resistance than chip resistors of the same type
- Products with lead-free terminations meet EU RoHS and China RoHS requirements

dimensions and construction



Vendor Part Number: CF1/4CT52R510J
Digikey Part Number: 2019-
CF1/4CT52R510JCT-ND - Cut Tape (CT)

Type	Dimensions inches (mm)				I**	
	L	C (max.)	D	d (nom.)	Standard	Long
CFS1/4 CFPS1/4	.126±.008 (3.2±0.2)	.134 (3.4)	.067 ^{+.008} _{-.004} (1.7 ^{+.02} _{-.01})	.018 (0.45)	.551 Min.* (14.0 Min.)	.787 Min.*** (20.0 Min.)
CF1/4	.240± ^{+.028} _{-.02}	.280	.092±.012	.024		

Datasheet Specs

- **51 Ω** value
- **¼ W** power rating
- **5 %** tolerance

Tolerance:

51 Ω with 5% tolerance

$$51\ \Omega \pm 0.05 * (51\ \Omega)$$

$$51\ \Omega \pm 2.55\ \Omega$$

$$48.45\ \Omega \leq \text{range} \leq 53.55\ \Omega$$

Power:

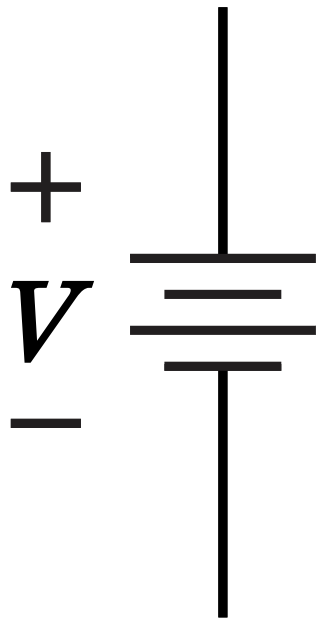
Operate below rating, ¼ W here

Parallel and Series Connections

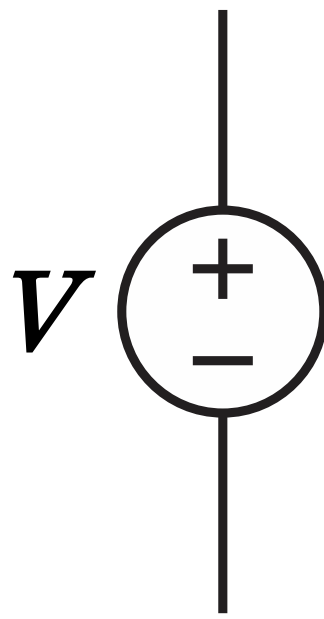


Review *Independent* Sources

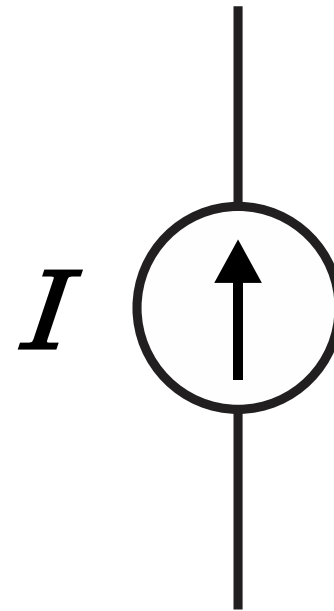
DC Voltage



(Battery)

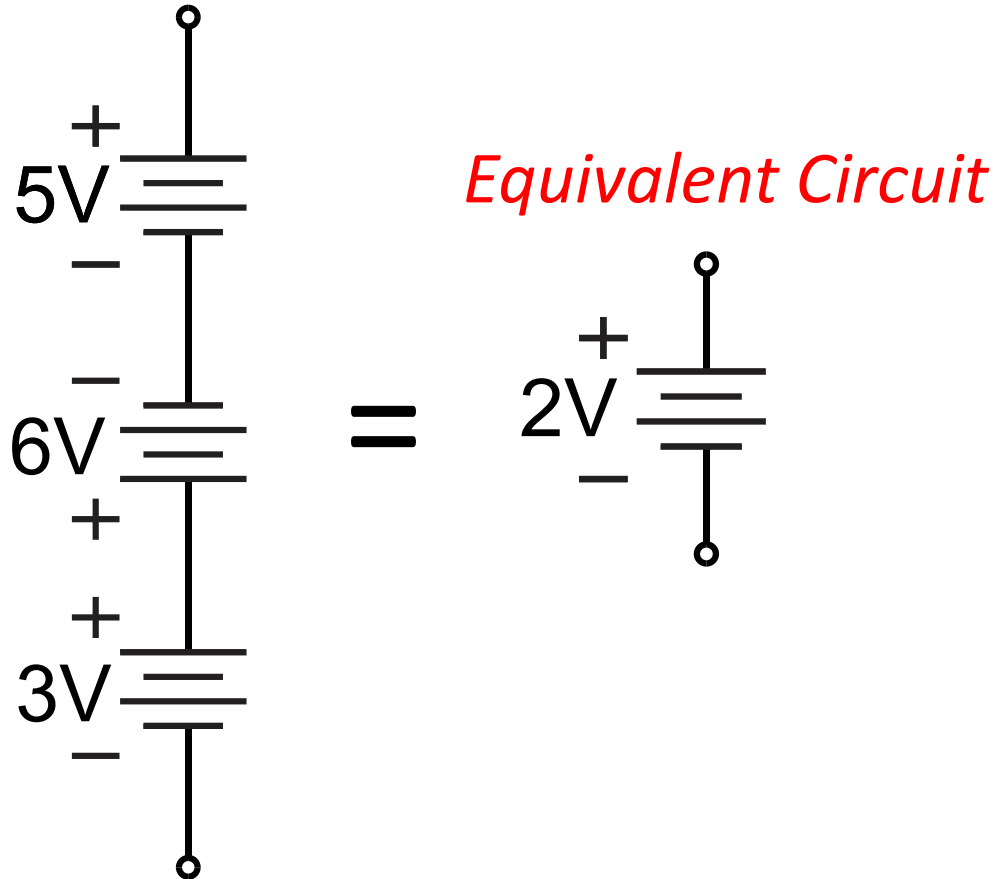


DC Current



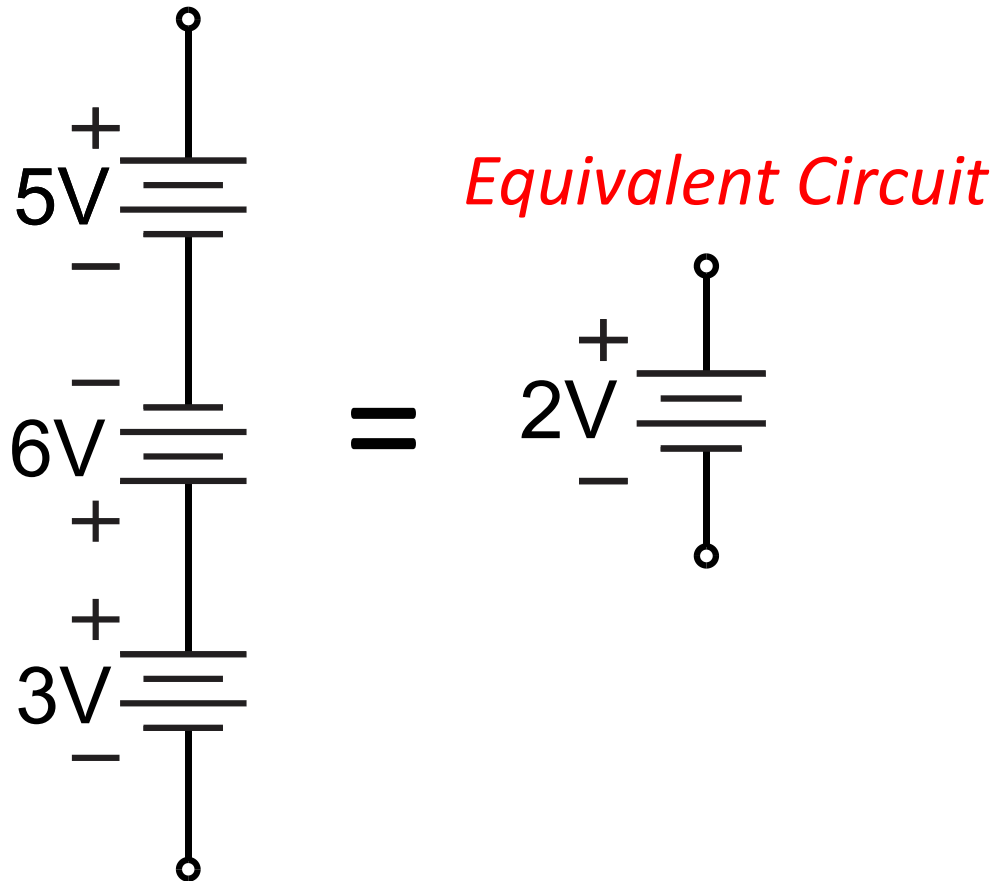
Equivalent Sources – Series Connected

Series-Connected Voltage Sources

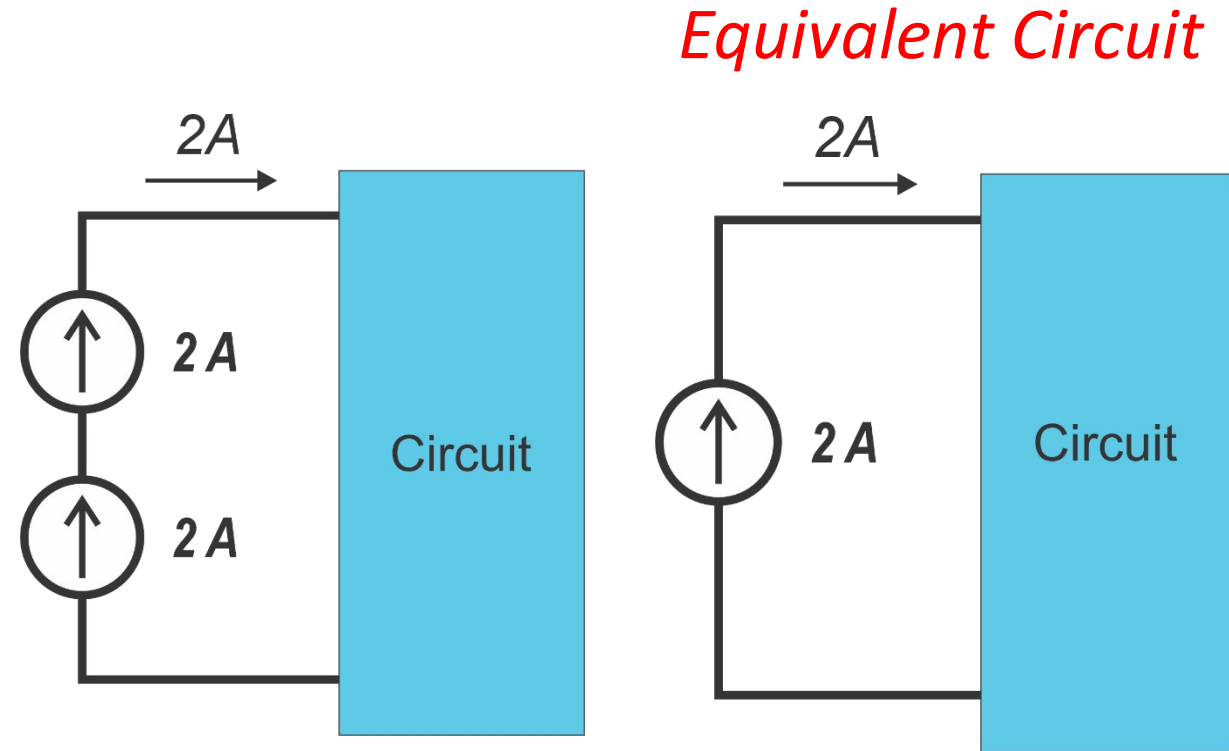


Equivalent Sources – Series Connected

Series-Connected Voltage Sources



Series-Connected Current Sources



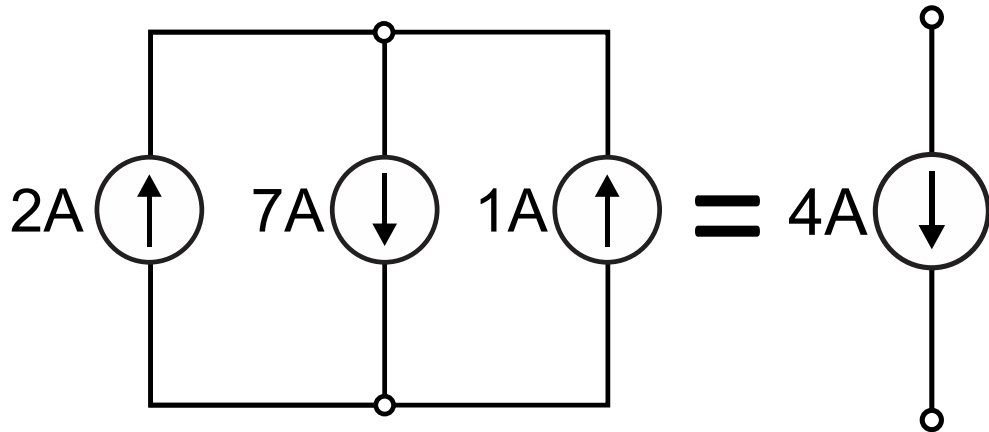
Series Connected Current Sources must have same value and direction



Equivalent Sources – Parallel Connected

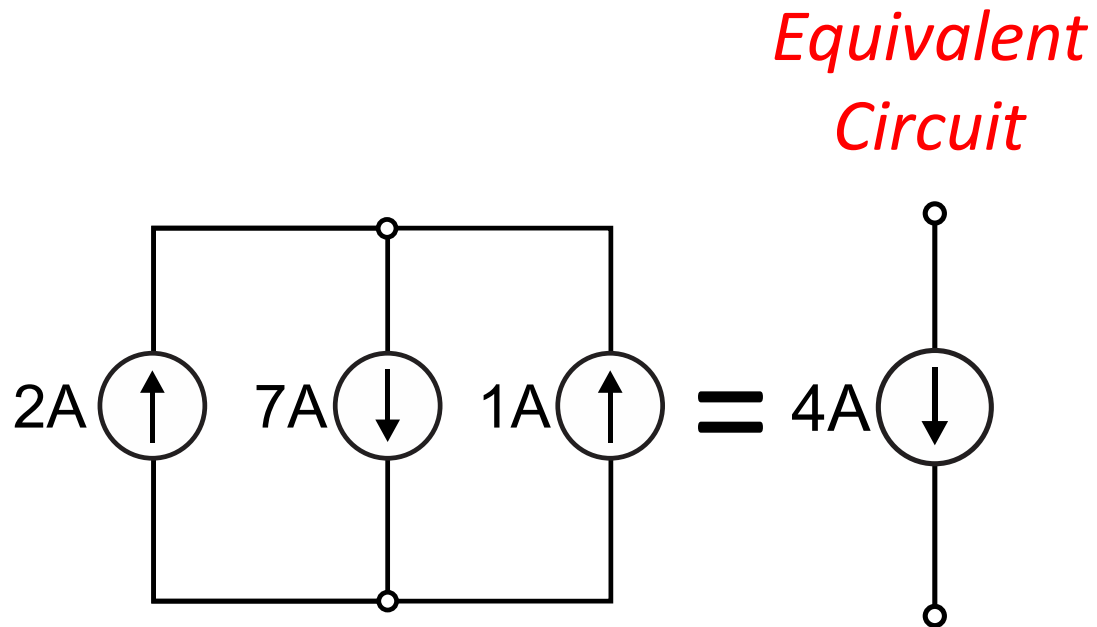
Parallel-Connected Current Sources

*Equivalent
Circuit*

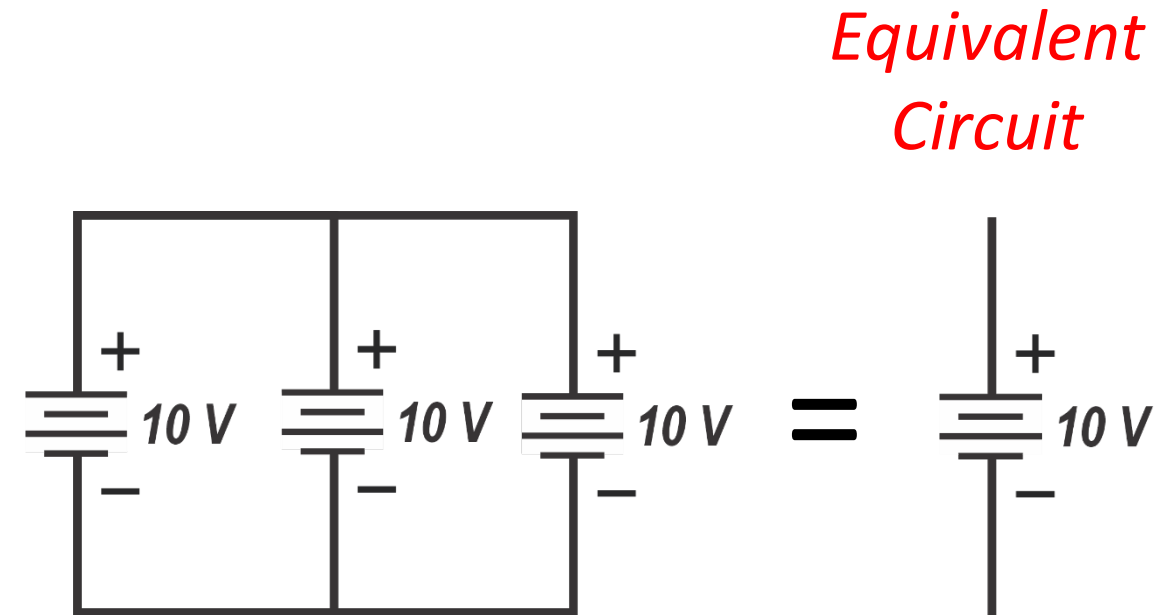


Equivalent Sources – Parallel Connected

Parallel-Connected Current Sources



Parallel-Connected Voltage Sources

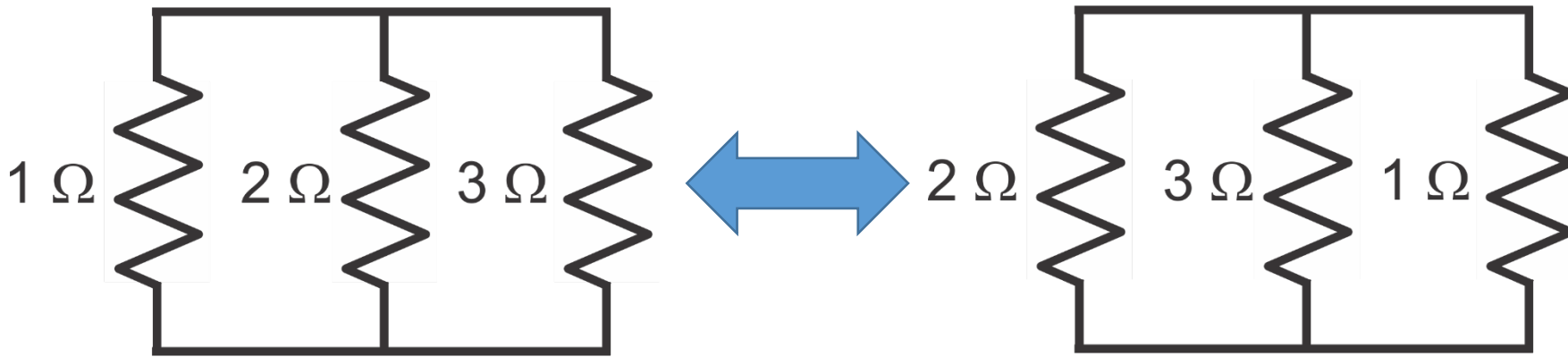


Parallel Connected Voltage Sources
must have same value and polarity



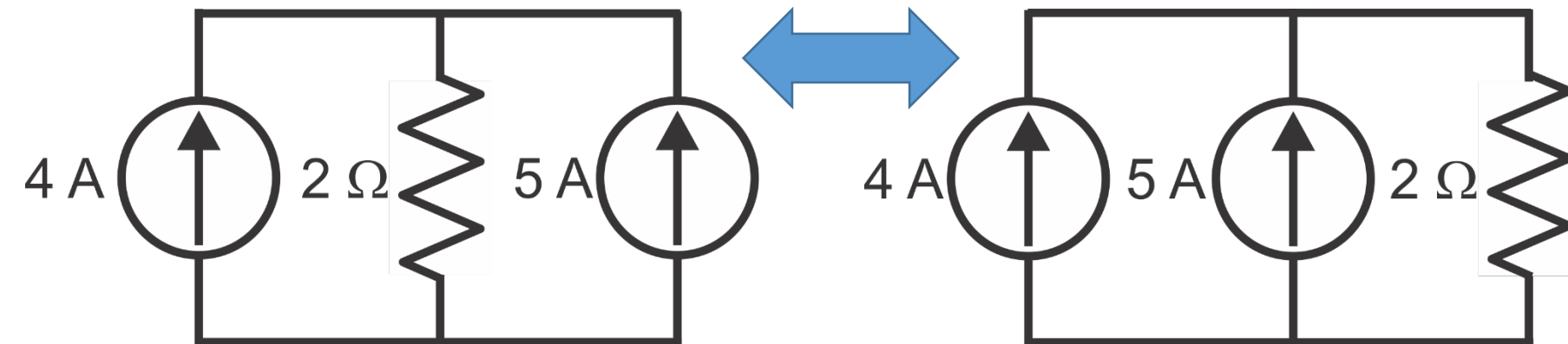
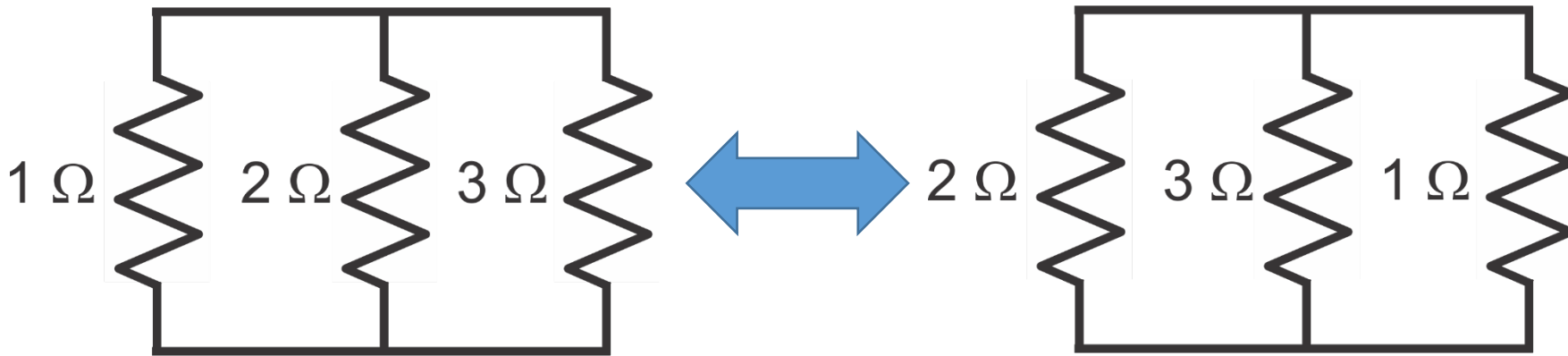
Parallel Components

Can be arranged in any order



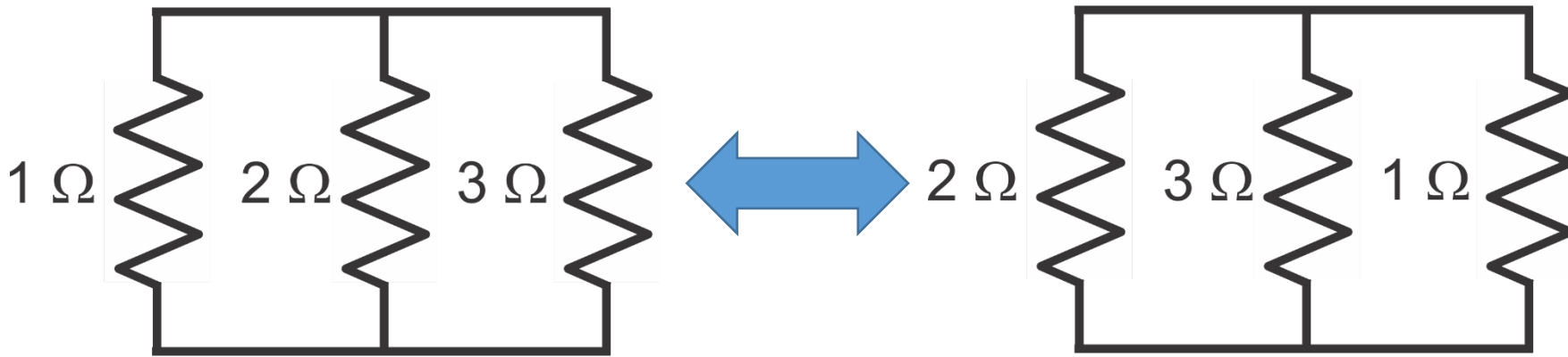
Parallel Components

Can be arranged in any order

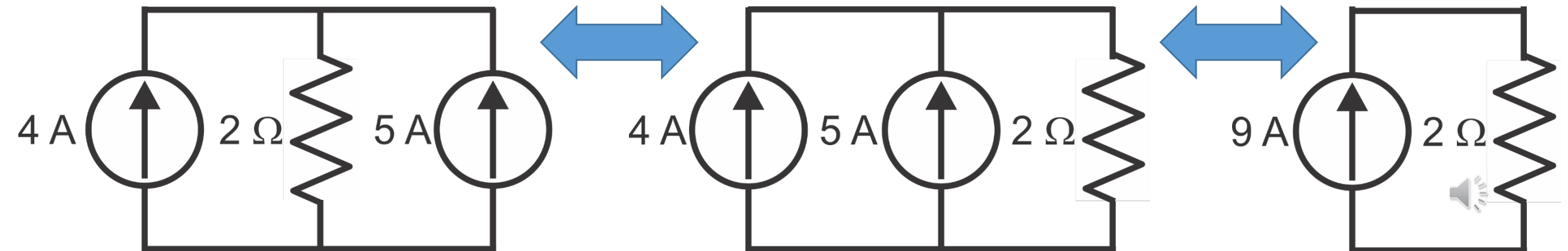


Parallel Components

Can be arranged in any order



Equivalent Circuit



Which is Correct?



Outline

- Ohm's Law
- Parallel and Series Components
- Kirchhoff's Law
- Ground
- Terminology of Short and Open Circuit



Kirchhoff's Current Law (KCL)

Kirchhoff's Voltage Law (KVL)

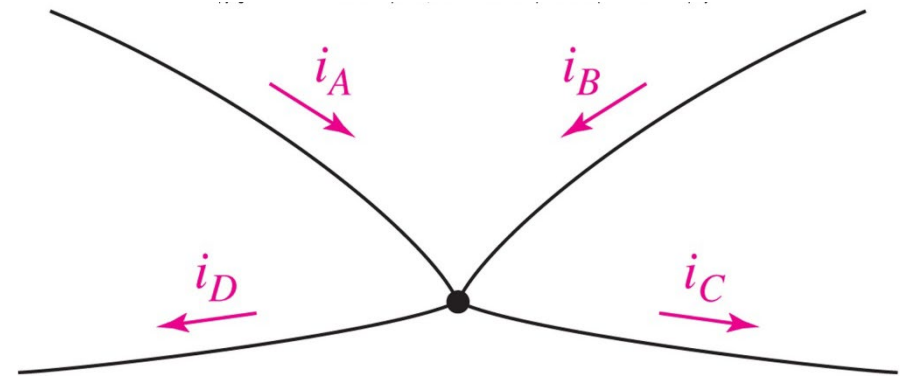


Kirchhoff's Current Law (KCL)

Current Law
(KCL)

Into Node

$$\sum_{n=1,2,\dots} I_n = 0$$



$$i_A + i_B + (-i_C) + (-i_D) = 0$$

Assign "in" direction as positive

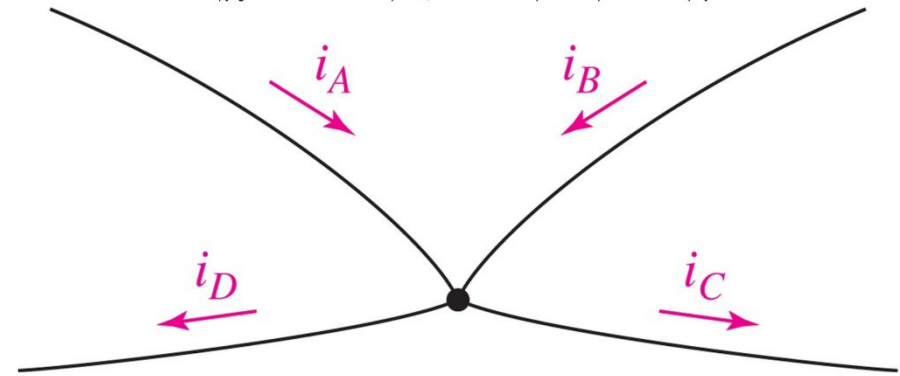


Kirchhoff's Current Law (KCL)

Current Law
(KCL)

Into Node

$$\sum_{n=1,2,\dots} I_n = 0$$



$$i_A + i_B + (-i_C) + (-i_D) = 0$$

$$i_A + i_B = i_C + i_D$$



IN



OUT

$$\sum_{n=1,2,\dots}^{\text{In}} I_n = \sum_{m=1,2,\dots}^{\text{Out}} I_m$$

KCL Summary

Into Node

$$\sum_{n=1,\dots} I_n = 0$$

Out of Node

$$\sum_{n=1,\dots} I_n = 0$$

$$\sum_{n=1,\dots}^{\text{In}} I_n = \sum_{m=1,\dots}^{\text{Out}} I_m$$



KCL Summary

Into Node

$$\sum_{n=1,\dots} I_n = 0$$

Out of Node

$$\sum_{n=1,\dots} I_n = 0$$

(most useful in Nodal Analysis)

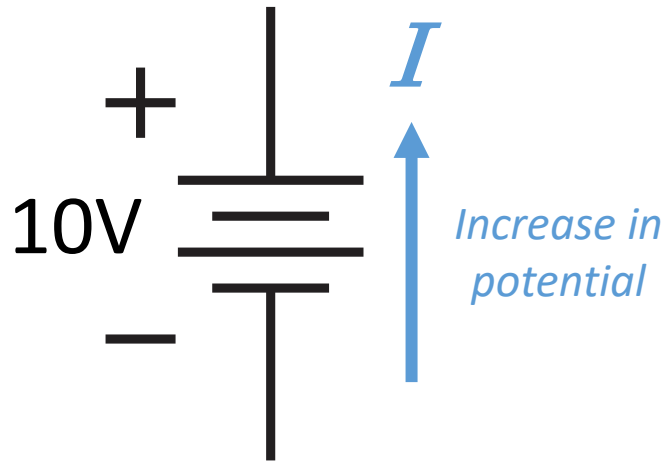
$$\sum_{n=1,\dots}^{\text{In}} I_n = \sum_{m=1,\dots}^{\text{Out}} I_m$$

(most intuitive)



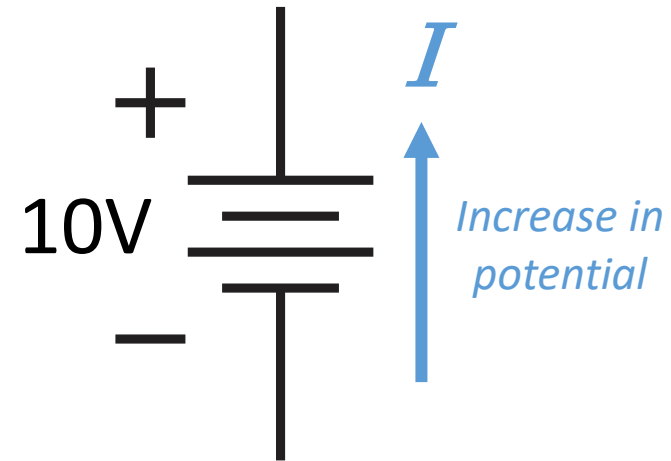
Voltage Rise or Voltage Drop?

Voltage Rise



Voltage Rise = +10 V

Voltage Drop



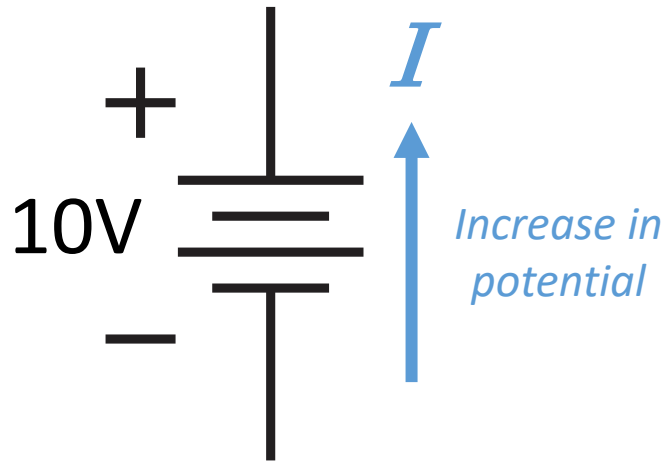
Voltage Drop = -10 V

(drop assumes a decrease in potential)



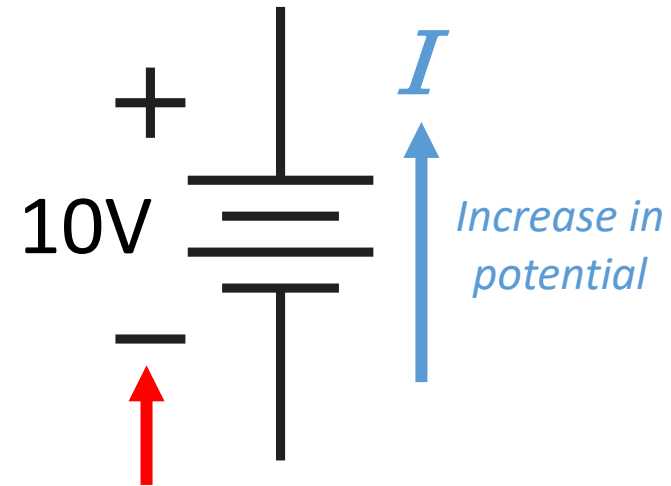
Voltage Rise or Voltage Drop?

Voltage Rise



Voltage Rise = +10 V

Voltage Drop



Easy to identify the drop by
the polarity first encountered

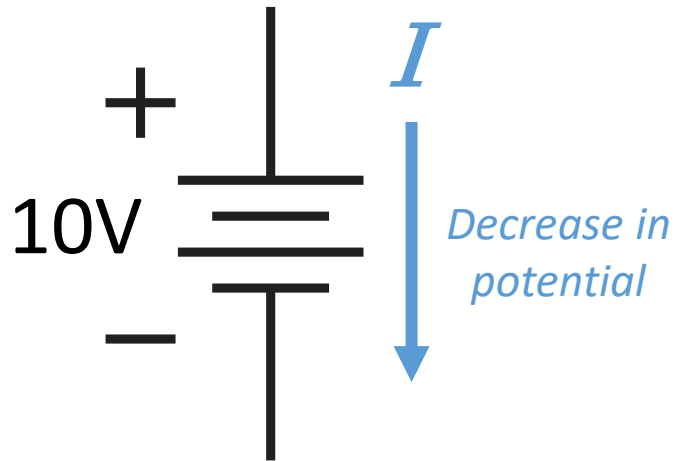
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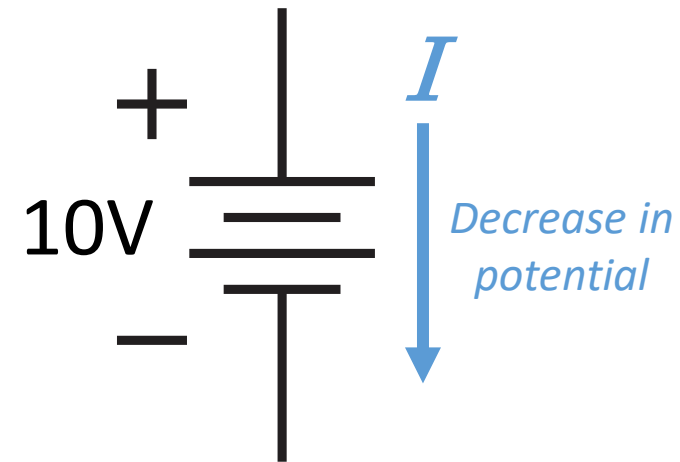
Voltage Rise or Voltage Drop?

Voltage Rise



Voltage Rise = -10 V

Voltage Drop



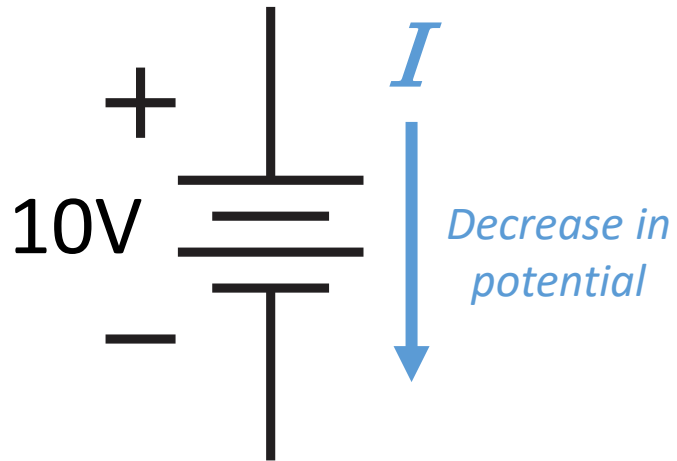
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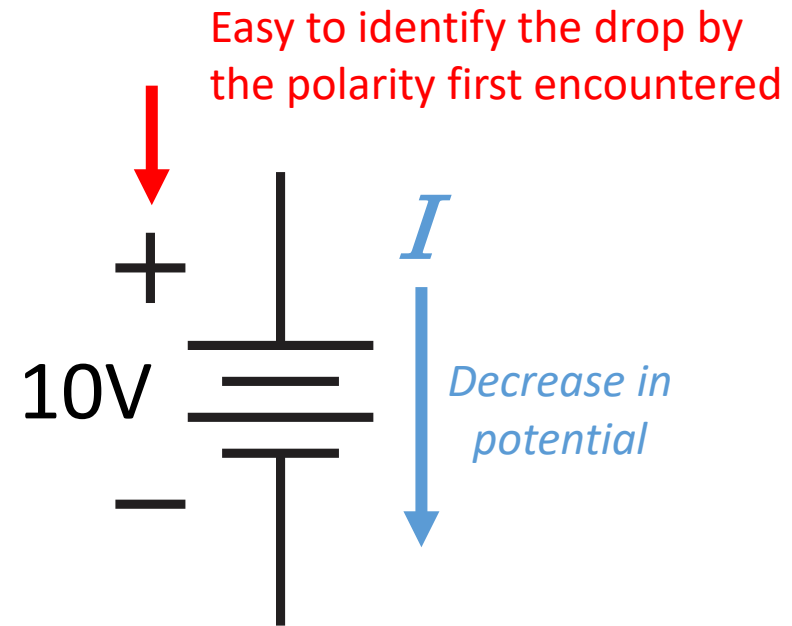
Voltage Rise or Voltage Drop?

Voltage Rise



Voltage Rise = -10 V

Voltage Drop



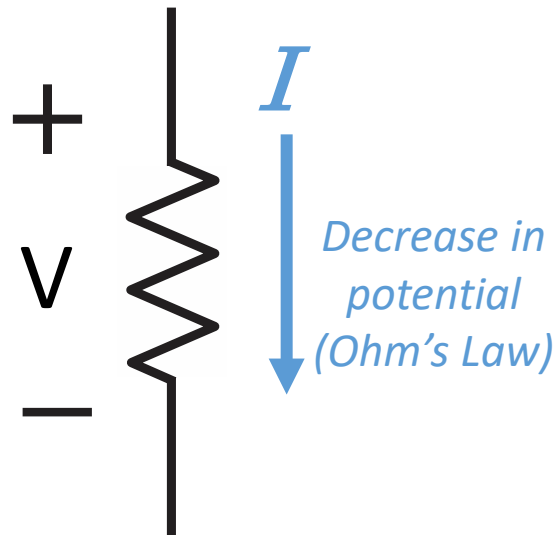
Voltage Drop = +10 V

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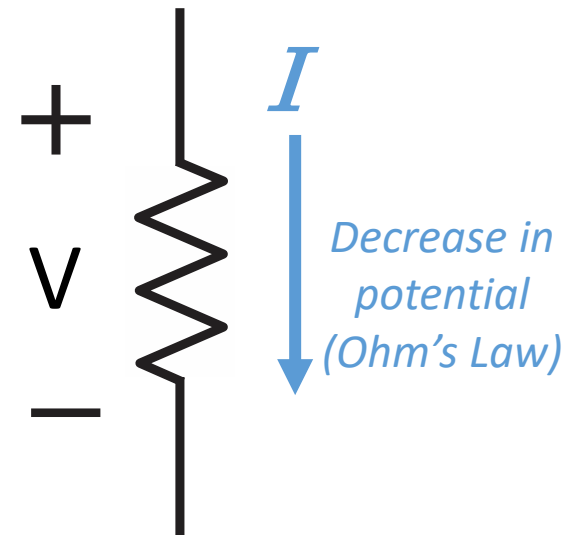
Voltage Rise or Voltage Drop?

Voltage Rise



Voltage Rise = -10 V

Voltage Drop



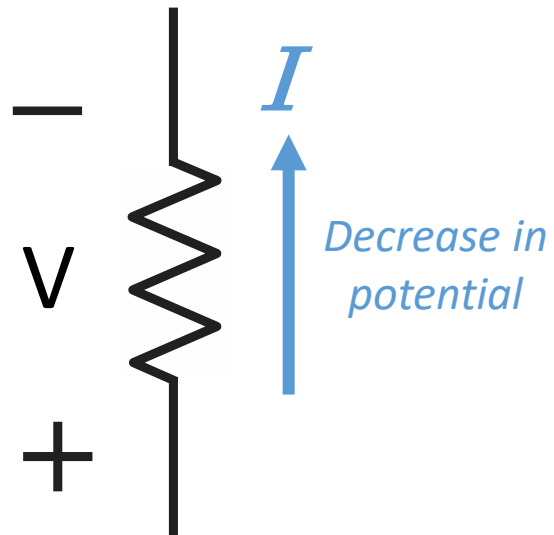
Voltage Drop = +10 V

(resistors always have a drop in the direction of current)



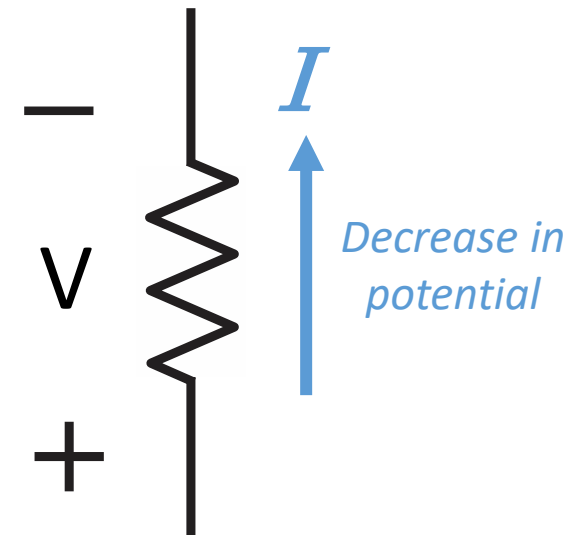
Voltage Rise or Voltage Drop?

Voltage Rise



Voltage Rise = -10 V

Voltage Drop



Voltage Drop = +10 V

(resistors always have a drop in the direction of current)



Kirchhoff's Voltage Law (KVL)

Around loop

$$\sum_{i=1, \dots} V_i = 0$$



Kirchhoff's Voltage Law (KVL)

Around loop

$$\sum_{i=1,\dots} V_i = 0$$

Which will you add?

Around loop

$$\sum_{i=1,\dots} V_{i, \text{rises}} = 0$$

Around loop

$$\sum_{i=1,\dots} V_{i, \text{drops}} = 0$$


Kirchhoff's Voltage Law (KVL)

Around loop

$$\sum_{i=1,\dots} V_i = 0$$

Which will you add?

Around loop

$$\sum_{i=1,\dots} V_{i, \text{rises}} = 0$$

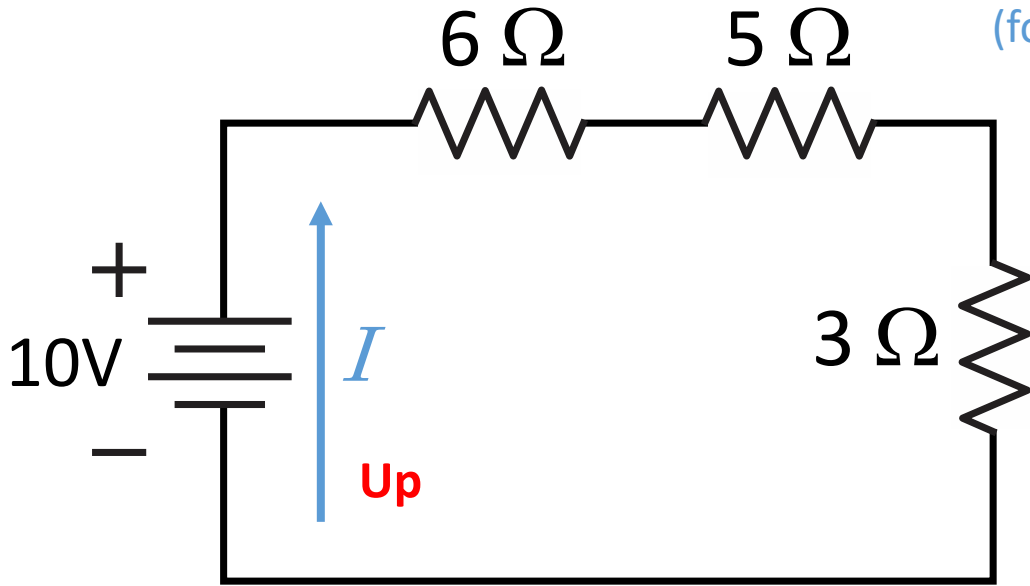
Around loop

$$\sum_{i=1,\dots} V_{i, \text{drops}} = 0$$


(Less Effort)

KVL Example

Assume direction for I
(follow loop in this direction)



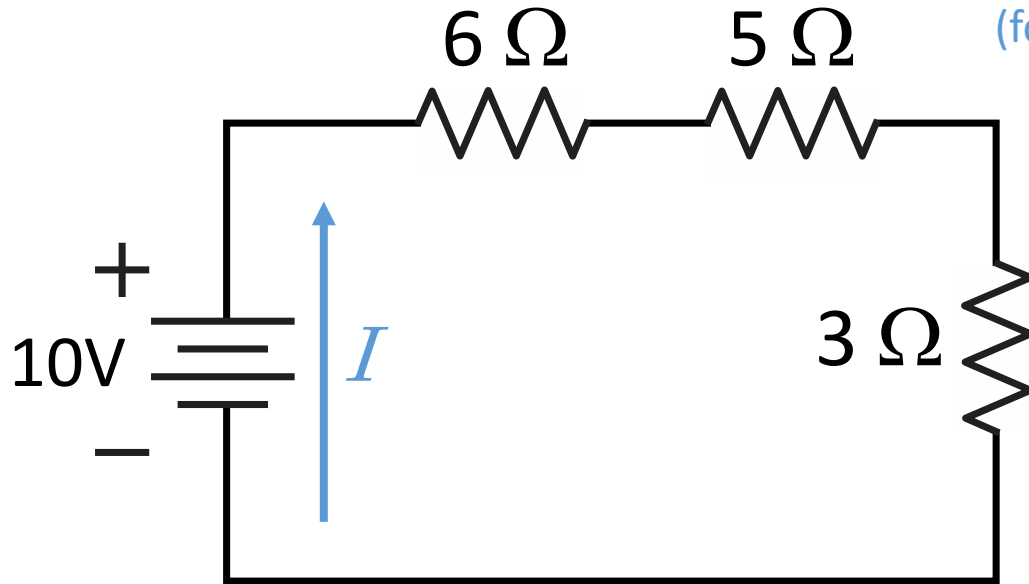
$$\sum_{i=1, \dots} V_{i, \text{rises}} = 0$$

$$+10 \text{ V} - I(6 \Omega) - I(5 \Omega) - I(3 \Omega) = 0$$

$$I = \frac{-10 \text{ V}}{-(6 \Omega + 5 \Omega + 3 \Omega)} = \frac{10 \text{ V}}{14 \Omega} = 714 \text{ mA}$$



KVL Example

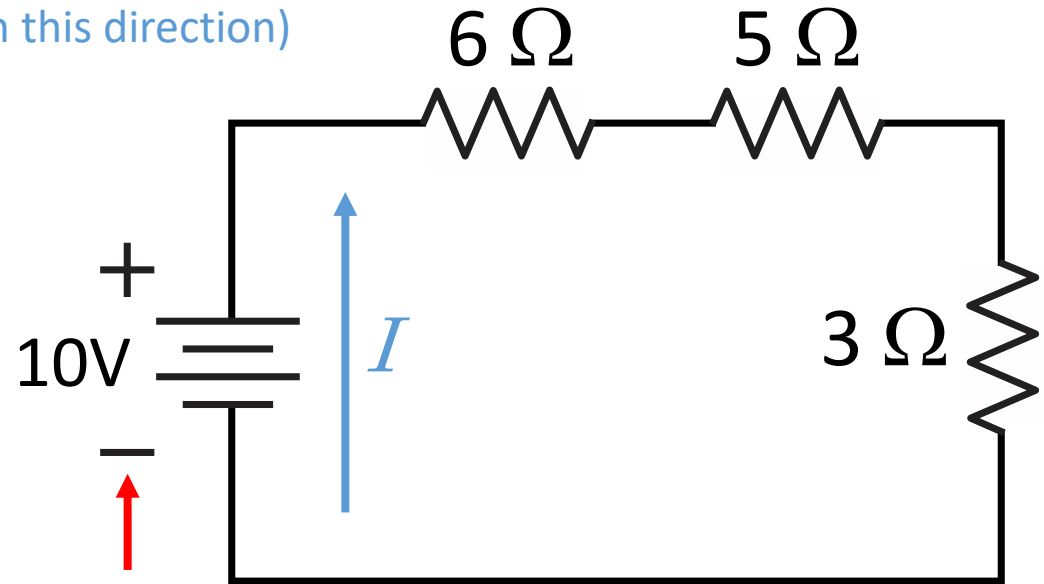


Assume direction for I
(follow loop in this direction)

$$\sum_{i=1, \dots} V_{i, \text{rises}} = 0$$

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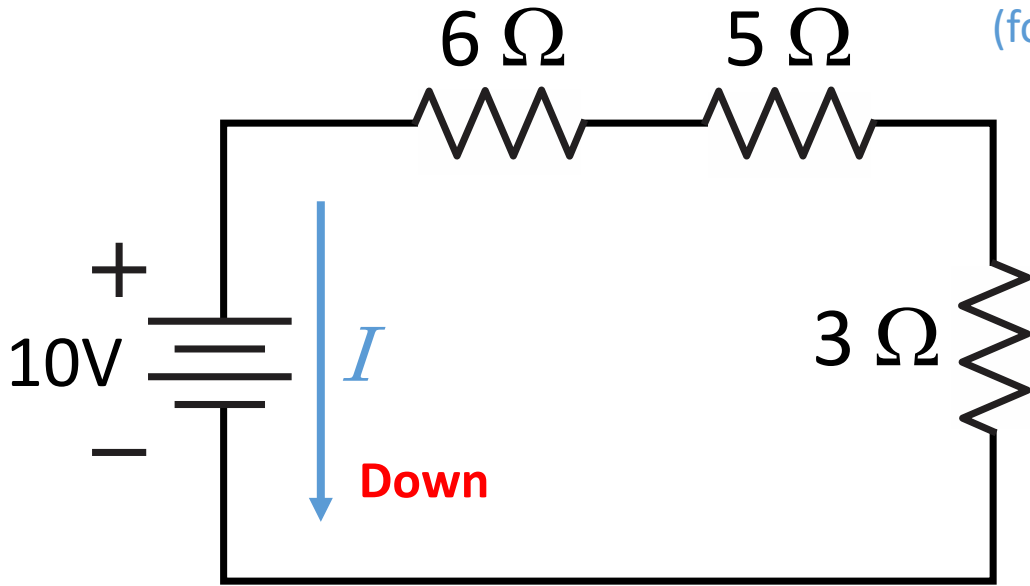
$$\sum_{i=1, \dots} V_{i, \text{drops}} = 0$$

$$-10 \text{ V} + I(6 \Omega) + I(5 \Omega) + I(3 \Omega) = 0$$

$$I = \frac{+10 \text{ V}}{(6 \Omega + 5 \Omega + 3 \Omega)} = \frac{10 \text{ V}}{14 \Omega} = 714 \text{ mA}$$

KVL Example

Assume direction for I
(follow loop in this direction)



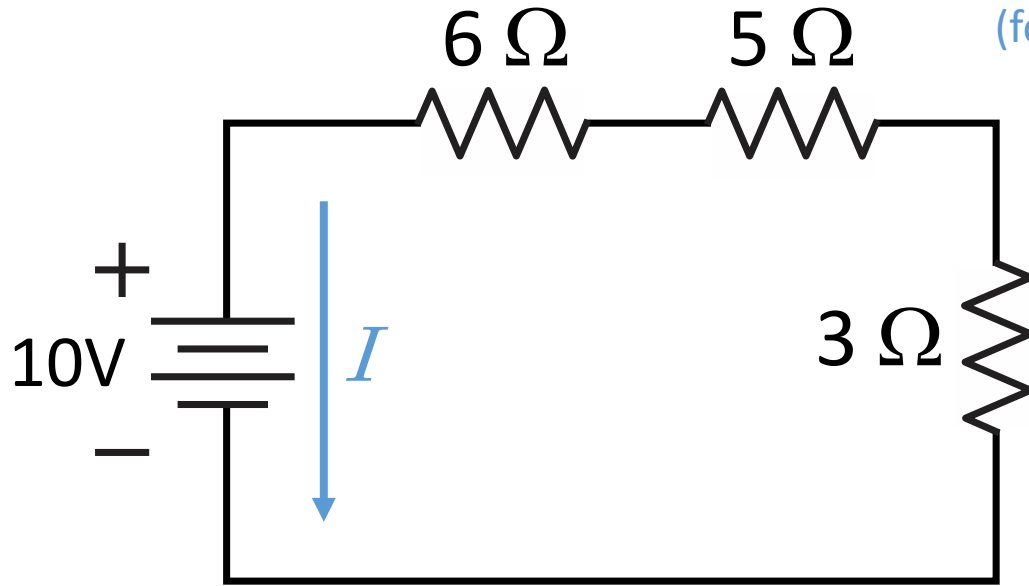
$$\sum_{i=1, \dots} V_{i, \text{ rises}} = 0$$

$$-10 \text{ V} - I(6 \Omega) - I(5 \Omega) - I(3 \Omega) = 0$$

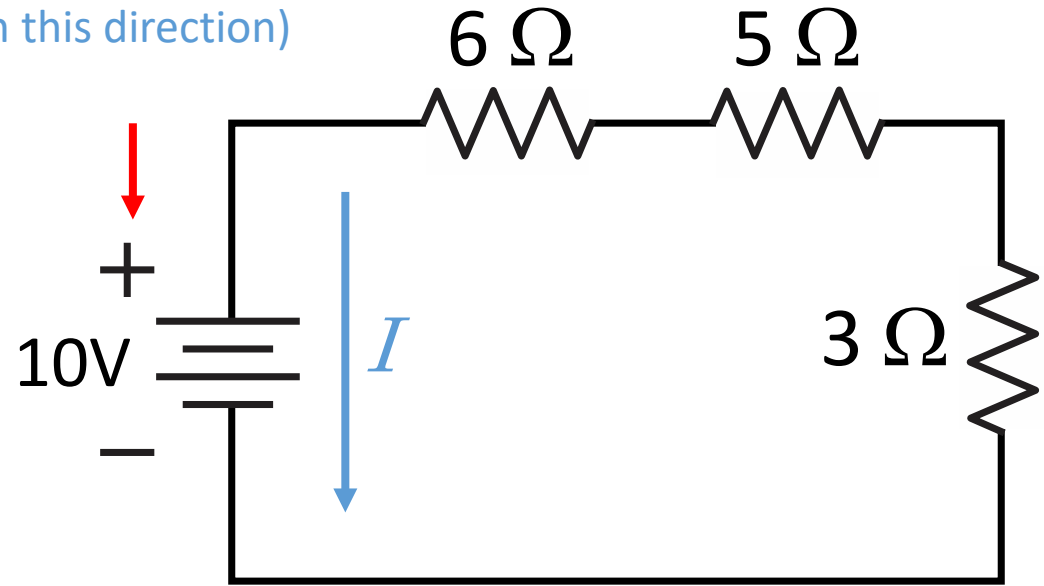
$$I = \frac{+10 \text{ V}}{-(6 \Omega + 5 \Omega + 3 \Omega)} = \frac{+10 \text{ V}}{-14 \Omega} = -714 \text{ mA}$$



KVL Example



Assume direction for I
(follow loop in this direction)



$$\sum_{i=1, \dots} V_{i, \text{rises}} = 0$$

$$-10 \text{ V} - I(6 \Omega) - I(5 \Omega) - I(3 \Omega) = 0$$

$$I = \frac{+10 \text{ V}}{-(6 \Omega + 5 \Omega + 3 \Omega)} = \frac{+10 \text{ V}}{-14 \Omega} = -714 \text{ mA}$$

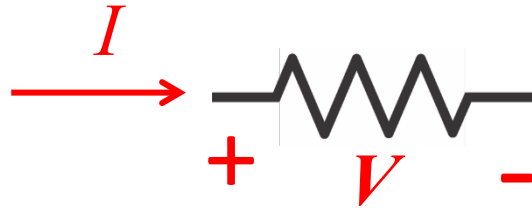
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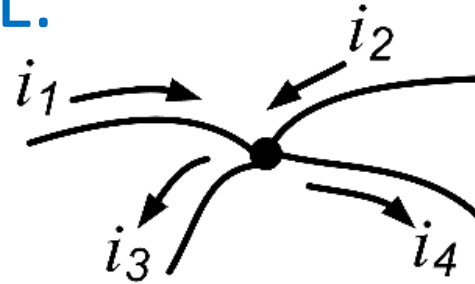
Circuits Toolbox

Ohm's Law:



$$V = IR$$

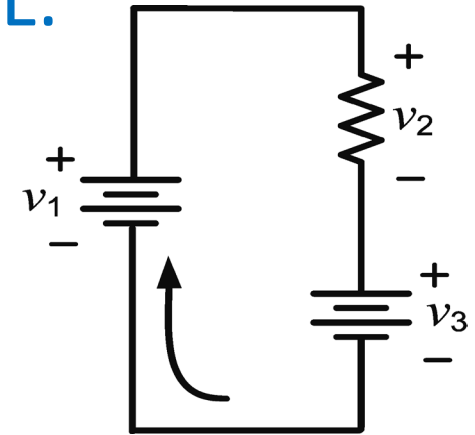
KCL:



$$\sum i_{\text{entering}} = \sum i_{\text{leaving}}$$

$$i_1 + i_2 = i_3 + i_4$$

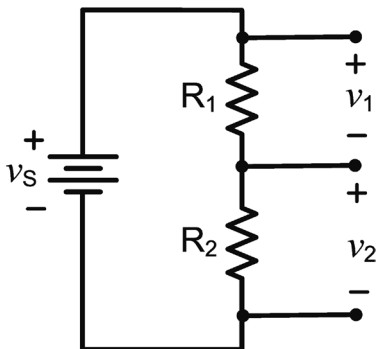
KVL:



$$\sum_{\text{loop}} v_i = 0$$

$$-v_1 + v_2 + v_3 = 0$$

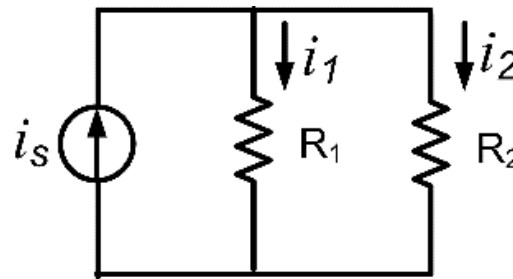
Voltage Divider:



$$v_1 = v_S \frac{R_1}{R_1 + R_2}$$

$$v_2 = v_S \frac{R_2}{R_1 + R_2}$$

Current Divider:



$$i_1 = i_S \frac{R_2}{R_1 + R_2}$$

$$i_2 = i_S \frac{R_1}{R_1 + R_2}$$

As circuits get more complicated, we need an organized method of applying Ohm's, KCL and KVL

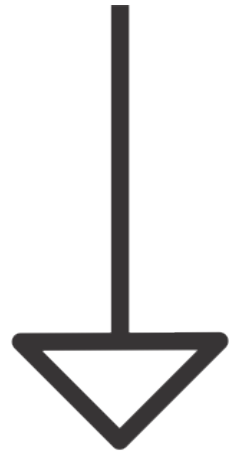
Electrical Symbols: Ground

Earth
Ground

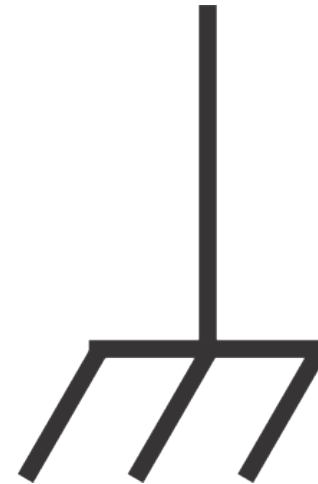


Used most often
as “ground”
(0 V reference)

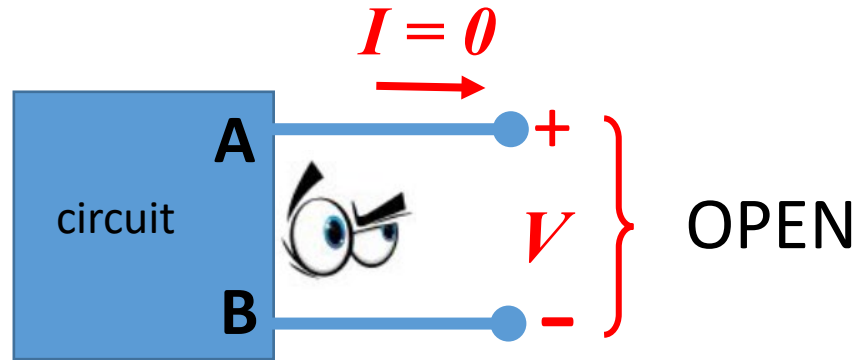
Common
Ground



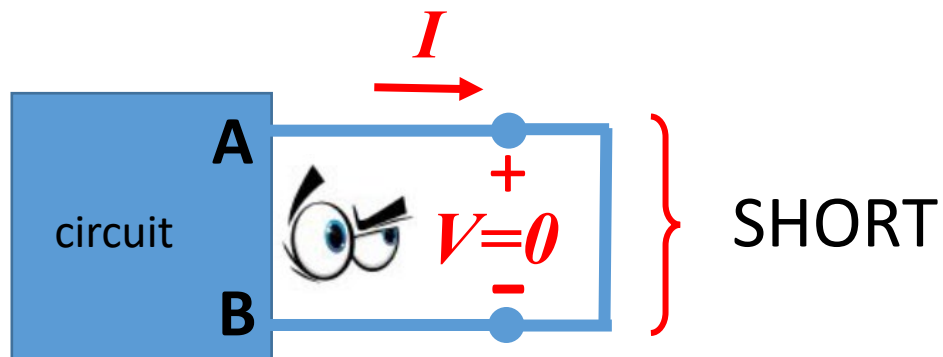
Chassis or
Frame



Terminology: Open and Short Circuits



- An open circuit between A and B means $i=0$
- *Voltage across* an open circuit: any value
- An open circuit is equivalent to $R = \infty \Omega$



- A short circuit between A and B means $v=0$
- *Current through* a short circuit: any value
- A short circuit is equivalent to $R = 0 \Omega$

